G H RAISONI COLLEGE OF ENGINEERING & MANAGEMENT, WAGHOLI, PUNE

(An Empowered Autonomous Institute, Affiliated to Savitribai Phule Pune University, Approved by AICT. New Delhi and Recognized by Govt. of Maharashtra.

(NAAC ACCREDIATED A+ Grade)

Department of Information Technology

Subject: Python Programming
[23UQCMDP2302]

Mini Project Report On

"Green Screen Replacement using Python"

Academic Year 2024-25

Semester-III

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Guided By:-

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Submitted by:

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4. **SITA57: Rucha Chaudhari**

GHraisoni COLLEGE
Engineering and Management
Pune

CERTIFICATE

This is to certify that **"**Green Screen Replacement**"** embodies the original work done by **during Miss Devyani Mendhe, Miss Renuka Nawale, Miss Riddhi Bagal, Miss Rucha Chaudhari** this project submission as a partial fulfilment of the requirement for the Mini Project using Python Programming subjects of Second Year B. Tech. Degree, III Semester, of Pune University during the academic year 2024-2025.

Date:

Place: Pune

(Mrs. Archana Dongardive	**(Dr. Poonam Gupta)**
Project Guide	**Head of Department**

ACKNOWLEDGEMENT

We would like to express our sincere thanks to **Mrs. Archana Dongardive** under whose valuable guidance and light of knowledge, we could complete this project.

We take this opportunity to thank all the staff members of Department Of Information Technology Engineering for their help whenever required. Finally we express sincere thanks to all those who have helped us directly or indirectlyin many ways in completion of this project work and I would like to extend my Deep appreciation to all my group members, without their support and Coordination we would not have been able to complete this Project.

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Project Statement: Green Screen Replacement in mp4 video

The goal of the project is to replace the **green screen with the background image in a video**, using **Python, OpenCv and NumPy.**

Key Points of the Program:

1. **Green Screen Detection:** The primary task is to detect the green screen by identifying a specific color range in **HSV(Hue, saturation,value).**

2. **Mask Creation:** Using trackbar values, lower and upper thresholds for the green screen are set using NumPy arrays. These thresholds are used to create Mask (**cv2.inRange()**) that detects green screen.

3. **Background Replacement:** The mask is inverted to isolate the green screen area, which is replaced by the corresponding area of new background image using **np.where()**.

4. **Color Space Conversion:** The video frames are converted from BGR to HSV color space using **cv2.cvtColor()**.

5. **Isolating the Subject:** The mask is Applied using **cv2.bitwise_and()** to isolate foreground (subject) from green screen background.

6. **Video Capture and Image Loading:** The video file is loaded using **cv2.VideoCapture()** and image is loaded for replacement using **cv2.imread().**

USES OF OpenCv AND NumPy:

1. **OpenCv:**
    a. **Video Capture and Frame Processing:** cv2.VideoCapture() loads video frame by frame. And Frame is processed in Loops.

    b. **Trackbars:** The Trackbars are created using cv2.createTrackbar() allow interactive tuning of green green detection.

    c. **Masking and Color Threshold:** cv2.Range() generates binary mask by identifying green pixels based of HSV range set by Trackbars.

    d. **Bitwise Operation :** cv2.bitwise_and() isolated subject by applying the mask and other bitwise operations replace green area with new background.

2. **NumPy:**
    a. **Array Manipulation:** NumPy arrays store pixel data, allowing efficient slicing and manipulation of the image regions.

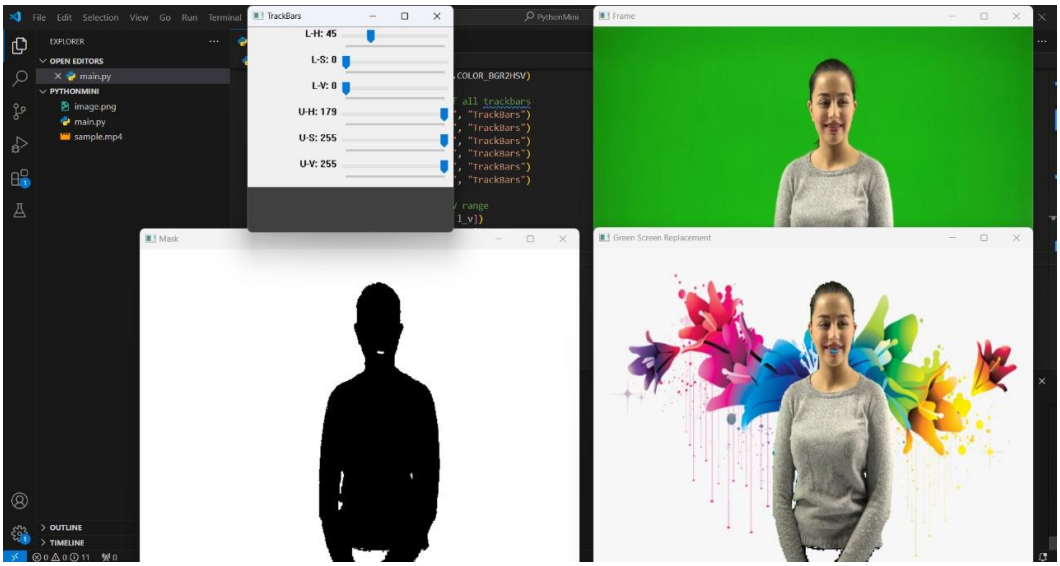
    b. **Green Screen Replacement:** np.where() is used to replace green screen pixel (where the mask is detected) with corresponding pixels from new image

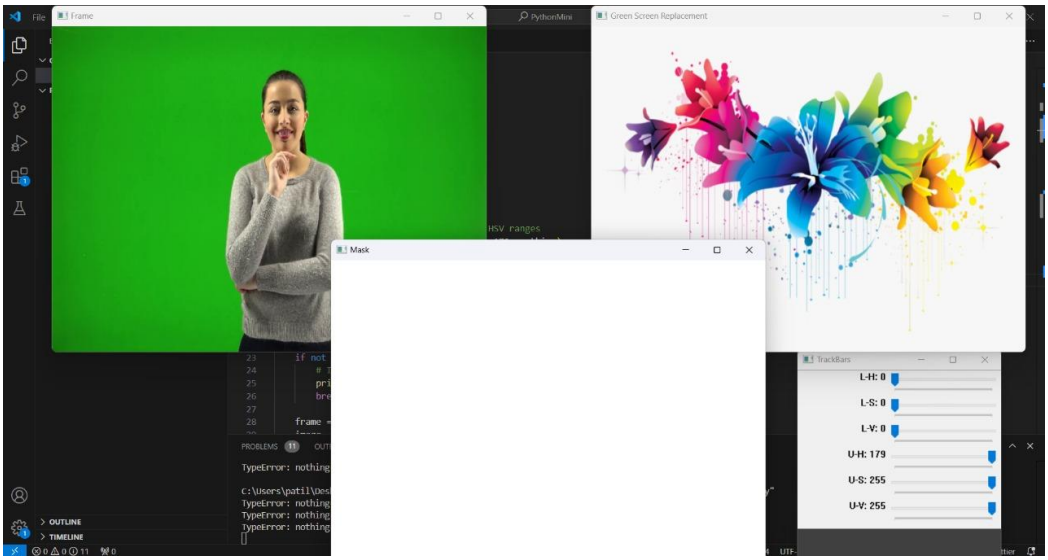
CODE

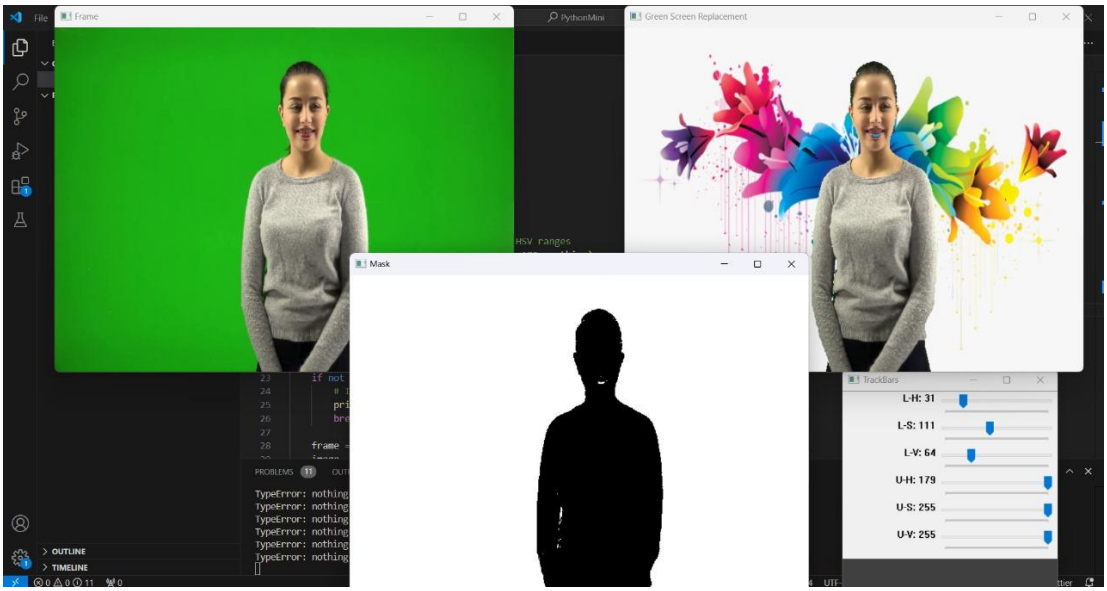
```
import cv2
import numpy as np

video = cv2.VideoCapture("sample.mp4")
image = cv2.imread("image.png")

def nothing():
    pass

cv2.namedWindow("TrackBars")
cv2.resizeWindow("TrackBars", 300, 300)

cv2.createTrackbar("L-H", "TrackBars", 0, 179, nothing)
cv2.createTrackbar("L-S", "TrackBars", 0, 255, nothing)
cv2.createTrackbar("L-V", "TrackBars", 0, 255, nothing)
cv2.createTrackbar("U-H", "TrackBars", 179, 179, nothing)
cv2.createTrackbar("U-S", "TrackBars", 255, 255, nothing)
cv2.createTrackbar("U-V", "TrackBars", 255, 255, nothing)

while True:
    ret, frame = video.read()
    if not ret:
        print("Video finished or not readable.")
        break

    frame = cv2.resize(frame, (640, 480))
    image = cv2.resize(image, (640, 480))
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

    l_h = cv2.getTrackbarPos("L-H", "TrackBars")
    l_s = cv2.getTrackbarPos("L-S", "TrackBars")
    l_v = cv2.getTrackbarPos("L-V", "TrackBars")
    u_h = cv2.getTrackbarPos("U-H", "TrackBars")
    u_s = cv2.getTrackbarPos("U-S", "TrackBars")
    u_v = cv2.getTrackbarPos("U-V", "TrackBars")

    l_green = np.array([l_h, l_s, l_v])
    u_green = np.array([u_h, u_s, u_v])

    mask = cv2.inRange(hsv, l_green, u_green)
    res = cv2.bitwise_and(frame, frame, mask=mask)
    f = frame - res
    green_screen = np.where(f == 0, image, f)

    cv2.imshow("Frame", frame)
    cv2.imshow("Mask", mask)
    cv2.imshow("Green Screen Replacement", green_screen)
```

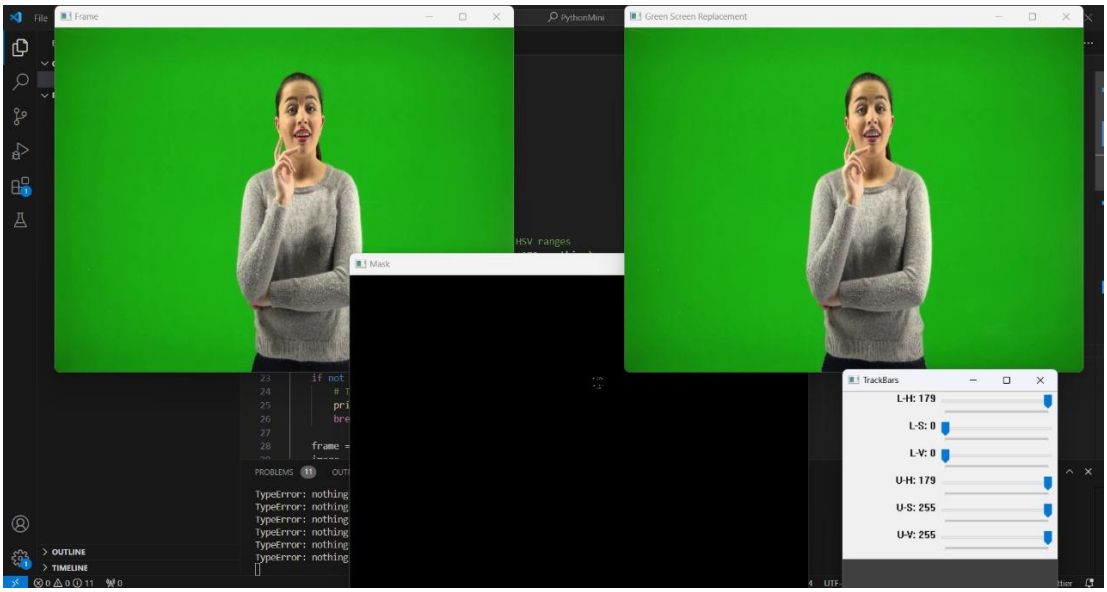
```
    k = cv2.waitKey(1)
    if k == ord('q'):
        break

print("Press any key to close the windows.")
cv2.waitKey(0)

video.release()
cv2.destroyAllWindows()
```

OUTPUT






Frame
PythonMini
Green Screen Replacement
Mask
TrackBars
L-H: 31
L-S: 111
L-V: 64
U-H: 179
U-S: 255
U-V: 255
HSV ranges
PROBLEMS
TypeError: nothing
OUTLINE
TIMELINE


Frame
PythonMini
Green Screen Replacement
Mask
TrackBars
L-H: 179
L-S: 0
L-V: 0
U-H: 179
U-S: 255
U-V: 255
HSV ranges
PROBLEMS
TypeError: nothing
OUTLINE
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Thank You !